

KAZYR, Russian Federation

WMO: 298760

Lat: 53.6861N

Lon: 94.2042E

Elev: 446

StdP: 96.08

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.3	-27.6	-34.0	0.2	-30.5	-30.4	0.2	-27.5	6.7	-15.2	4.9	-12.7	0.1	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	30.1	19.0	28.0	18.0	25.9	17.2	20.4	26.9	19.4	25.1	18.4	23.7	1.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	14.1	22.7	17.5	13.2	21.5	16.6	12.5	20.3	61.2	27.1	57.5	25.1	54.2	23.6	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.9	3.6	3.1	-34.4	34.0	4.6	2.3	-37.7	35.6	-40.5	37.0	-43.1	38.3	-46.4	40.0		
(5)			-35.2	22.3	4.3	1.5	-38.3	23.3	-40.8	24.2	-43.2	25.0	-46.3	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-13.6	-10.8	-3.1	3.5	8.9	16.0	18.5	16.8	10.1	2.4	-7.1	-11.2		
	DBStd	12.25	6.28	6.21	5.84	4.42	4.62	4.48	2.91	3.42	4.39	4.53	6.52	6.12		
	HDD10.0	3463	731	583	406	197	79	6	0	1	51	239	514	656		
	HDD18.3	5828	989	816	664	443	294	97	34	70	247	494	764	914		
	CDD10.0	767	0	0	0	4	44	184	263	213	56	3	0	0		
	CDD18.3	90	0	0	0	0	1	26	38	24	1	0	0	0		
	CDH23.3	1122	0	0	0	3	59	426	381	223	30	0	0	0		
	CDH26.7	331	0	0	0	0	11	143	109	62	6	0	0	0		
(14) Wind	WSAvg	0.8	0.9	0.8	0.9	1.0	0.9	0.8	0.7	0.6	0.7	0.8	1.0	0.9		
(15) Precipitation	PrecAvg	760	29	25	34	50	67	86	106	109	77	66	65	47		
	PrecMax	920	48	75	77	84	116	128	180	189	140	193	112	95		
	PrecMin	602	9	8	10	21	18	42	52	64	26	21	38	13		
	PrecStd	80	10	13	17	18	23	24	25	26	27	31	18	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.8	5.0	14.6	21.6	28.0	32.7	31.7	31.4	27.0	18.8	7.6	1.2		
		MCWB	-2.5	0.0	6.1	10.8	14.7	19.9	19.9	20.5	15.9	10.5	4.5	-0.4		
	2%	DB	-2.6	1.4	10.6	17.7	24.9	30.1	29.8	28.2	22.7	14.6	4.6	-0.5		
		MCWB	-4.0	-1.5	3.9	8.9	13.9	18.5	19.2	19.3	14.9	8.7	2.3	-2.0		
	5%	DB	-3.7	-0.6	7.5	14.7	21.7	28.1	27.5	25.6	19.7	11.0	1.8	-2.0		
		MCWB	-5.1	-2.4	2.2	7.4	11.7	17.8	18.4	18.0	13.7	6.6	0.0	-3.4		
	10%	DB	-5.3	-2.4	4.9	11.6	18.3	25.5	25.3	23.3	16.7	8.4	0.3	-3.3		
		MCWB	-6.8	-4.2	0.9	5.6	10.8	16.5	17.9	17.1	12.2	5.6	-1.0	-4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.9	0.8	6.6	11.9	16.5	22.4	21.7	21.5	17.1	11.5	5.1	0.5		
		MCDB	-1.3	3.2	13.0	18.5	26.1	30.6	28.4	28.1	23.5	17.4	6.9	1.3		
	2%	WB	-3.6	-0.8	4.4	9.6	14.6	20.1	20.2	20.1	15.5	9.3	2.4	-1.9		
		MCDB	-2.8	0.9	9.7	16.6	22.1	26.6	26.9	26.0	21.4	13.7	4.1	-1.0		
	5%	WB	-5.1	-2.3	2.7	7.7	13.1	18.6	19.4	18.9	14.3	7.6	0.4	-3.1		
		MCDB	-3.8	-0.6	6.9	13.3	19.8	25.2	25.0	24.2	19.0	10.5	1.5	-2.2		
	10%	WB	-6.8	-4.4	1.2	6.3	11.4	17.5	18.5	17.8	12.8	5.9	-0.9	-4.3		
		MCDB	-5.4	-2.5	4.5	11.2	16.9	24.1	23.5	21.9	16.0	8.1	0.2	-3.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	11.4	12.6	11.8	12.6	13.7	11.4	11.2	10.7	8.1	6.6	7.3		
		MCDBR	7.3	11.6	15.3	17.3	19.6	18.9	16.4	16.1	16.3	13.9	7.2	6.3		
		MCWBR	6.3	8.5	9.4	9.5	9.6	8.0	6.8	7.4	8.3	8.3	5.2	5.7		
		MCDBR	6.8	10.1	13.9	16.1	17.2	16.2	13.7	14.3	15.2	12.7	6.1	5.8		
	5% WB	MCWBR	6.0	7.4	8.6	9.2	9.4	7.6	6.6	7.1	7.9	8.0	4.8	5.5		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.336	0.383	0.426	0.433	0.409	0.418	0.407	0.373	0.359	0.336	0.305		
	taud		1.985	1.929	1.864	1.933	2.101	2.314	2.318	2.316	2.339	2.194	1.977	1.925		
	Ebn at Noon		614	713	764	791	814	842	825	812	793	696	546	499		
	Edh at Noon		78	116	155	165	148	121	119	112	97	89	79	69		
(44) All-Sky Solar Radiation	RadAvg		1.03	2.04	3.36	4.38	5.01	5.74	5.23	4.35	3.00	1.70	0.90	0.68		
	RadStd		0.10	0.17	0.26	0.23	0.50	0.63	0.40	0.38	0.39	0.17	0.13	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page